ULTIMATE 2026 GENAI VAULT — 100+ QUESTIONS

TOP 100+ GENAI INTERVIEW QUESTIONS

100+ Complete AI & LLM Engineering Questions

Master Transformers, Prompting, Embeddings, Vector DBs, RAG, Fine-Tuning, AI Agents, Evaluation & System Design.

100+ Vault | Transformers | Prompting | Embeddings | Vector DBs | RAG ☆ | Fine-Tuning | AI Agents ☆ | System Design ☆

SWIPE TO MASTER ALL 100+ QUESTIONS →

1. LLM & TRANSFORMER FUNDAMENTALS

100+ Vault: Q 01 - 20

01 What is a Large Language Model (LLM)?

02 How does a Transformer architecture work?

03 Why were Transformers preferred over RNNs and LSTMs?

04 What is self-attention?

05 How does Multi-Head Attention work?

06 What is the Query-Key-Value (Q, K, V) mechanism?

07 What is positional encoding and why is it needed?

08 What is the difference between encoder-only, decoder-only, and encoder-decoder Transformers?

09 Why are models like GPT decoder-only?

10 What is causal/self-attention masking?

11 What is tokenization?

12 How do BPE, WordPiece, and SentencePiece differ?

13 What is a context window?

14 What happens when an input exceeds the context window?

15 What is perplexity in language models?

16 What are model parameters?

17 What is the difference between parameters, tokens, and embeddings?

18 What is temperature in LLM generation?

19 What are Top-K and Top-P (nucleus) sampling?

20 How do temperature, Top-K, and Top-P affect model output?

2. PROMPT ENGINEERING

21 What is prompt engineering?

22 What is zero-shot prompting?

23 What is few-shot prompting?

24 What is chain-of-thought (CoT) prompting?

25 What is role prompting?

26 What is prompt decomposition?

27 What is structured output generation?

28 How do you make an LLM reliably return JSON?

29 What are prompt injection attacks?

30 How do you defend an LLM application against prompt injection?

3. EMBEDDINGS & VECTOR DATABASES

31 What are embeddings?

32 How are text embeddings generated?

33 What is the difference between embedding models and LLMs?

34 What is cosine similarity?

35 What is the difference between cosine similarity, Euclidean distance, and dot product?

36 How do you choose an embedding model?

37 What is a vector database?

38 How does vector similarity search work?

39 What is ANN (Approximate Nearest Neighbor) search?

40 How do HNSW and IVF indexes work?

41 What factors affect vector search quality?

42 What is metadata filtering in vector search?

4. RETRIEVAL-AUGMENTED GENERATION (RAG)

100+ Vault: Q 43 - 60

43 What is Retrieval-Augmented Generation (RAG)?

44 Why use RAG instead of fine-tuning?

45 Explain the end-to-end RAG pipeline.

46 What is document chunking?

47 How do you choose chunk size and chunk overlap?

48 What are the different chunking strategies?

49 What is semantic chunking?

50 What problems occur with naive chunking?

51 What is hybrid search?

52 How does BM25 differ from vector search?

53 What is reranking, and why is it useful in RAG?

54 Why is reranking useful in RAG?

55 How would you design a production-grade RAG system?

56 How do you handle tables and images in RAG?

57 How do you handle very large documents?

58 How do you reduce irrelevant retrieval?

59 How do you prevent hallucinations in RAG?

60 How do you evaluate a RAG system?

61 When should you fine-tune an LLM vs use RAG?

62 When should you use RAG instead of fine-tuning?

63 What is supervised fine-tuning (SFT)?

64 What is LoRA (Low-Rank Adaptation)?

65 What is QLoRA, and how does it optimize memory?

66 What is PEFT (Parameter-Efficient Fine-Tuning)?

67 What is the difference between full fine-tuning and LoRA?

68 What is quantization?

69 What are 4-bit and 8-bit quantization (GPTQ / AWQ / GGUF)?

70 What is catastrophic forgetting and how do you mitigate it?

71 How do you prepare a dataset for fine-tuning?

72 How do you evaluate a fine-tuned model?

73 What is an AI agent?

74 How is an AI agent different from a chatbot?

75 What is tool/function calling in LLMs?

76 How does an LLM decide when to call a tool?

77 What is an agentic workflow?

78 What is the difference between deterministic workflows and autonomous agents?

79 What is the ReAct (Reason + Act) framework?

80 How do agents maintain memory and state?

81 How do you prevent an agent from entering infinite loops?

82 How do you design reliable multi-agent systems?

83 What are the major failure modes of AI agents?

84 How do you evaluate an AI agent?

85 How do you implement human-in-the-loop (HITL) for agents?

7. GENAI EVALUATION

86 Why is evaluating LLM applications difficult?

87 What is LLM-as-a-Judge?

88 What are hallucination evaluation techniques?

89 What are Precision@K, Recall@K, and MRR in RAG?

90 What are faithfulness, relevance, and context precision?

91 How would you create an evaluation dataset for a RAG application?

92 How do you perform offline vs online evaluation?

8. PRODUCTION & LLM SYSTEM DESIGN

93 How would you design a production-grade GenAI application?

94 How do you reduce LLM latency?

95 How do you reduce LLM inference cost?

96 How do you handle LLM rate limits and failures?

97 How do you implement caching in an LLM application?

98 How do you monitor an LLM application in production?

99 How would you design a scalable RAG system for millions of documents?

100 Design an end-to-end production AI agent/RAG system & explain trade-offs, security, evaluation, cost & observability.

Topic Hierarchy for GenAI Interviews

Prioritize studying topics in this exact sequential order to maximize interview performance:

```
Transformers → Prompting → Embeddings → RAG → Reranking → Fine-tuning → Agents → Evaluation → LLMOps → System Design
```

High-Value Interviewer Favorite Questions (100+ List)

For Senior AI/ML & GenAI Engineer roles, focus deeply on these core numbers:

- ✓ #45 RAG Pipeline
- ✓ #47 Chunking Overlap
- ✓ #51 Hybrid Search
- ✓ #53 Reranking
- ✓ #55 Production RAG
- ✓ #59 Prevent Hallucinations
- ✓ #60 RAG Evaluation
- ✓ #61 Fine-tuning vs RAG
- ✓ #65 QLoRA Architecture
- ✓ #75 Function Calling
- ✓ #78 Agent Workflows
- ✓ #81 Agent Loops
- ✓ #83 Agent Failures
- ✓ #86 LLM Evaluation
- ✓ #93 GenAI Design
- ✓ #95 Cost Reduction
- ✓ #98 LLM Monitoring
- #100 End-to-End System

INTERVIEWER'S #1 FAVORITE QUESTION

Question 100+

"Design a production-grade GenAI application from scratch."

Be prepared to articulate all 12 core building blocks in your architectural answer:

1. Data & Preprocessing
PDF/HTML parsing, cleaning, metadata tag extraction.

2. Chunking Strategy
Semantic chunking, sliding windows, overlap ratio.

3. Embedding Engine
Domain fine-tuning, dimension choice, batching.

4. Vector Database
HNSW vs IVF indexing, filtering, sharding.

5. Hybrid Retrieval
Dense Vector Search + Sparse BM25 (RRF fusion).

6. Cross-Encoder Rerank
BGE-Reranker/Cohere reranking for top precision.

7. Prompt Engineering
System instructions, zero/few-shot context injection.

8. LLM & Tool Calling
Structured outputs, function calling, agent loop.

9. Guardrails & Safety
NeMo Guardrails, input/output prompt injection filters.

10. Caching & Latency
Exact/Semantic caching (GPTCache), streaming responses.

11. Evaluation Pipeline
RAGAS framework (Faithfulness, Context Relevance).

12. LLMOps & Cost
LangSmith/Phoenix tracing, token budgets, rate-limits.

LEVEL UP YOUR AI CAREER

SAVE & REPOST FOR YOUR NEXT INTERVIEW!

Follow Resnur AI on LinkedIn & Instagram for daily AI system design breakdowns, LLM cheatsheets, & technical interview prep!

Resnur AI

@resnur.ai